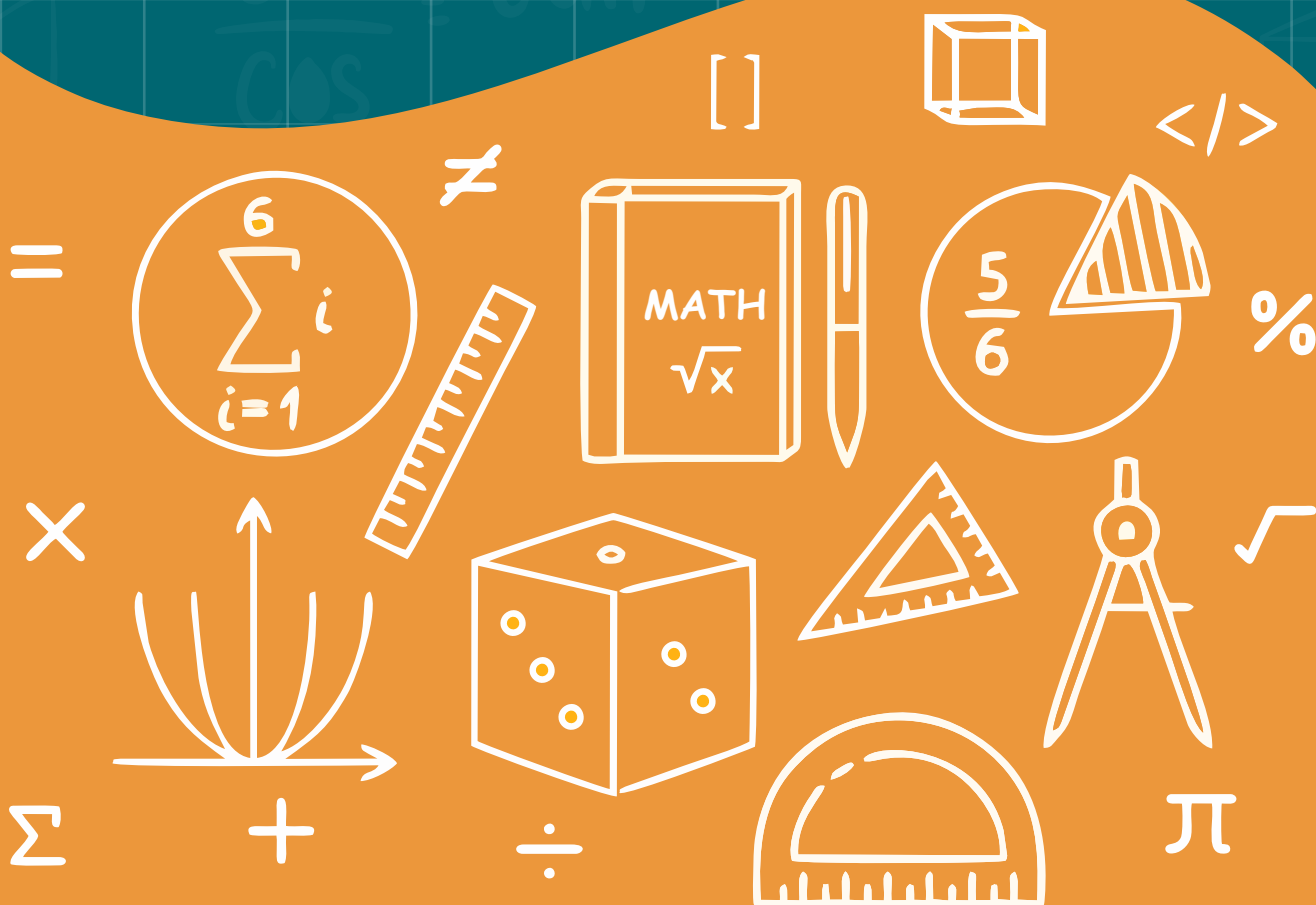


UNIT

3

FACTORS AND MULTIPLES



FACTORS

A factor is a number which divides the dividend completely leaving no remainder.

Example: Which of the following are the factors of 10.

(a) 2

(b) 4

(c) 5

(d) 6

Solutions:

(a)

$$\begin{array}{r} 2 \overline{) 10} \quad 5 \\ - 10 \\ \hline 0 \end{array}$$

$10 = 2 \times 5$

Since Remainder = 0
So, 2 is a factor of 10

(c)

$$\begin{array}{r} 5 \overline{) 10} \quad 2 \\ - 10 \\ \hline 0 \end{array}$$

$10 = 5 \times 2$

Since Remainder = 0
So, 5 is a factor of 10
So, 2 and 5 are the factors of 10

(b)

$$\begin{array}{r} 4 \overline{) 10} \quad 2 \\ - 8 \\ \hline 2 \end{array}$$

Since Remainder is not zero
So, 4 is not a factor of 10

(d)

$$\begin{array}{r} 6 \overline{) 10} \quad 1 \\ - 6 \\ \hline 4 \end{array}$$

Since Remainder is not zero
So, 6 is not a factor of 10

Example: Find all the factors of.

(a) 12

(b) 20

Solutions:

(a) 12

As

$$12 = 1 \times 12$$

$$12 = 2 \times 6$$

$$12 = 3 \times 4$$

So, 1, 2, 3, 4, 6, and 12 can divide 12 exactly.

Hence, 1, 2, 3, 4, 6, and 12 are factors of 12

(b) 20

As

$$20 = 1 \times 20$$

$$20 = 2 \times 10$$

$$20 = 4 \times 5$$

So, 1, 2, 4, 5, 10, and 20 can divide 20 exactly.

Hence, 1, 2, 4, 5, 10, and 20 are factors of 20.

MULTIPLES

Define a multiple as a dividend into which a factor can divide:

Let us consider the division of 18 by 2.

$$\begin{array}{r} 2 \overline{) 18} \quad 9 \\ \underline{- 18} \\ 0 \end{array}$$

$$18 = 2 \times 9$$

Here the divisor 2 is called a factor of 18 because the remainder is zero.

Multiple of a given number is the dividend, for which the given number is its factor.

So, a number has always unlimited number of multiples.

Example 1: Is 20 a multiple of 3?

Solutions:

$$\begin{array}{r} 3 \overline{) 20} \quad 6 \\ - 18 \\ \hline 2 \end{array}$$

Since 3 is not a factor of 20
So, 20 is not a multiple of 3.

Example 2: Find all multiples of 2.

Solutions:

Multiples of 2 can be found as:

$$2 \times 1 = 2$$

$$2 \times 2 = 4$$

$$2 \times 3 = 6$$

$$2 \times 4 = 8$$

and so on

So, the multiples of 2 are 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, ...

Example 3: Find the multiples of 3 less than 15.

Solutions: The multiples of 3 are 3, 6, 9, 12, 15, 18, ...

So, the multiples less than 15 are 3, 6, 9, 12.

EVEN NUMBERS

Rule: A number having 0, 2, 4, 6, or 8 at its ones place is an even number.

Example: Which of the following numbers are even:

- (a) 257 (b) 7208
(c) 1114 (d) 15683

Solutions:

- (a) 257 is not even because digit at ones place is 7.
(b) 7208 is even because digit at ones place is 8.
(c) 11114 is even because digit at ones place is 4.
(d) 15683 is not even because digit at ones place is 3.

ODD NUMBERS

An odd number is a number which is not exactly divisible by 2.

Rule: A number having 1, 3, 5, 7 or 9 at its ones place is an odd number.

PRIME NUMBERS

A natural number which has only two distinct factors 1 and number itself is called a prime number.

As 2, 3, 5, 7, 11, 13, 17, 19, ... have only two factors. So, they are prime numbers.

Example: Which of the following numbers is a prime number?

- (a) 23 (b) 8

Solutions:

- (a) 23 has two factors 1 and 23. So, 23 is a prime number.
(b) 8 has four factors 1, 2, 4, 8. So, 8 is not a prime number.

COMPOSITE NUMBERS

A natural number which has more than two factors is called a composite number.

As 4, 6, 8, 9, 10, 12, 14, 15, ... have more than two factors.
So, they are composite numbers.

Example: Identify the composite or prime numbers with reasons.

(a) 8 (b) 17

(c) 9 (d) 5

Solutions:

	Number	Type	Reasons
a	8	Composite number	As 1, 2, 4, 8 are the factors of 8, which are more than two. Hence it is a composite number.
b	17	Prime number	As 1, 17 are the only two distinct factors, so 17 is a prime number.
c	9	Composite number	As 1, 3, 9 are the factors of 9, which are more than two. Hence it is a composite number.
d	5	Prime number	As 1, 5 are the only two distinct factors of 5, so 5 is a prime number.

EXERCISE 3.1

Q1. Which of the following are the factors of 20.

(a) 2 (b) 8 (c) 5 (d) 3

Q2. Write all the factors of each of the following.

(a) 15 (b) 30 (c) 50 (d) 125 (e) 150

Q3. Which of the following are multiples of 6.

(a) 10 (b) 18 (c) 96 (d) 200

Q4. Write first seven multiples of:

(a) 4 (b) 7 (c) 50 (d) 15 (e) 20

Q5. Write prime numbers between:

(a) 6 and 15 (b) 20 and 40 (c) 60 and 95

Q6. Which of the following is even or odd?

- (a) 10 (b) 47 (c) 75 (d) 91 (e) 100 (f) 117

TEST FOR DIVISIBILITY

Test of Divisibility by 2:

A number is exactly divisible by 2 if the digit at units place of the number is 0, 2, 4, 6 or 8.

Example: 50, 62, 724, 9126, 3338 are exactly divisible by 2.

Test of Divisibility by 3:

A number is exactly divisible by 3 if the sum of the digits of the number is divisible by 3.

Example: 24510 is exactly divisible by 3 because $2 + 4 + 5 + 1 + 0 = 12$ and 12 is divisible by 3

Test of Divisibility by 4:

A number is exactly divisible by 4 if its last two digits are divisible by 4.

Example: 26720 is exactly divisible by 4 because 20 is divisible by 4.

Test of Divisibility by 5:

A number is divisible by 5 if the digit at the units place of the number is 0 or 5.

Example: 5210 and 4115 are exactly divisible by 5.

Test of Divisibility by 6:

A number is divisible by 6 if the number is exactly divisible by both 2 and 3.

Example: Example: 2142 is divisible by 6 because it is exactly divisible by both 2 and 3.

Test of Divisibility by 8:

A number is exactly divisible by 8 if the number formed by the last three digits is exactly divisible by 8 or the last three digits are zero.

Example: 213832 is divisible by 8 because 832 is exactly divisible by 8.

Test of Divisibility by 9:

A number is exactly divisible by 9 if the sum of the digits of the number is exactly divisible by 9.

Example: 9241011 is exactly divisible by 9 because $9 + 2 + 4 + 1 + 0 + 1 + 1 = 18$ and 18 is exactly divisible by 9

Test of Divisibility by 10:

A number is exactly divisible by 10 if the digit at units place of the number is zero.

Example: 25670 is exactly divisible by 10.

Example: Test whether:

- (i) 2412006 is exactly divisible by 3
- (ii) 6123816 is exactly divisible by 8
- (iii) 61710 is exactly divisible by 6

Solution:

(i) $2 + 4 + 1 + 2 + 0 + 0 + 6 = 15$. Since 15 is divisible by 3.
So, 2412006 is divisible by 3.

(ii) 6123816 is divisible by 8. Since 816 is divisible by 8.

(iii) $6 + 1 + 7 + 1 + 0 = 15$. Since 15 is divisible by 3 and digit at unit's place is 0, so it is divisible by 2. Hence 61710 is divisible by 6.

EXERCISE 3.2

Q1. Test whether.

- (i) 251061 is exactly divisible by 3
- (ii) 92348 is exactly divisible by 4
- (iii) 49230 is exactly divisible by 9
- (iv) 24150 is exactly divisible by 8
- (v) 10005 is exactly divisible by 2
- (vi) 241566 is exactly divisible by 3
- (vii) 34268 is exactly divisible by 2

Q2. Which of the following numbers are exactly divisible by 6?

- (i) 2456 (ii) 7121700 (iii) 19206

Q3. Which of the following numbers are exactly divisible by 5?

- (i) 24567 (ii) 230590 (iii) 111165

Q4. Which of the following numbers are exactly divisible by 10?

- (i) 25670 (ii) 123600 (iii) 14675

Q5. Which of the following numbers are exactly divisible by 2, 3, and 6?

- (i) 14562 (ii) 1101361 (iii) 518214

Q6. Which of the following numbers are exactly divisible by 4 and 8?

- (i) 513864 (ii) 617116 (iii) 2415848

Q7. Write the first five multiples for each number.

(i) 6 _____, _____, _____, _____, _____

(ii) 8 _____, _____, _____, _____, _____

(iii) 12 _____ / _____ / _____ / _____ / _____

(iv) 15 _____ / _____ / _____ / _____ / _____

(v) 50 _____ / _____ / _____ / _____ / _____

FACTORIZATION

The process of factorizing a number into its factors is called factorization.

For example:

(i) $8 = 2 \times 4$

Here 2 and 4 are factors of 8

(ii) $12 = 6 \times 2$

Here 6 and 2 are factors of 12

Prime Factorization:

The process of factorizing a number into its prime factors is called prime factorization.

For example:

(i) $8 = 2 \times 2 \times 2$

(ii) $12 = 2 \times 2 \times 3$

Example 1: Find the prime factorization of 60.

Solution:

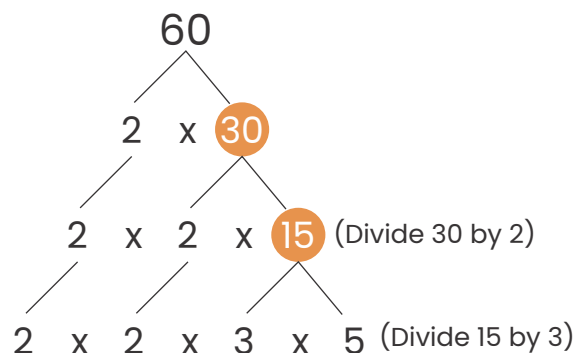
Division method

2	60
2	30
3	15
5	5
	1

So, $60 = 2 \times 2 \times 3 \times 5$

which is prime factorization of 60

Tree method



So, $60 = 2 \times 2 \times 3 \times 5$

which is prime factorization of 60

Example 2: Find the prime factorization of 540.

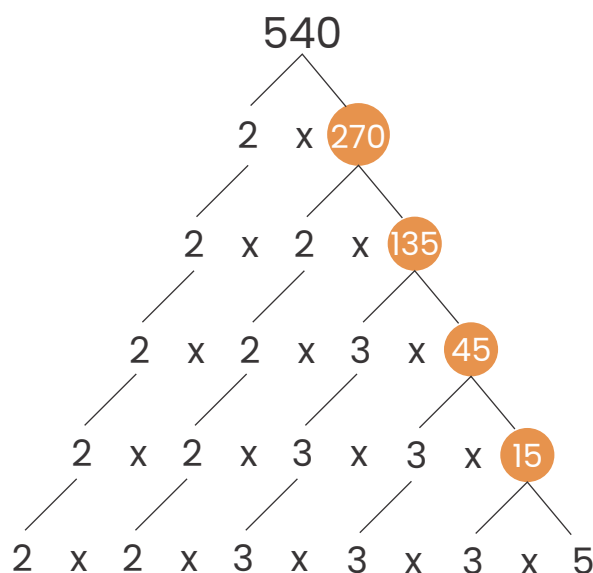
Solution:

Division method

2	540
2	270
3	135
3	45
3	15
5	5
	1

So, $540 = 2 \times 2 \times 3 \times 3 \times 3 \times 5$
which is prime factorization of 540.

Tree method



So, $540 = 2 \times 2 \times 3 \times 3 \times 3 \times 5$
which is prime factorization of 540.

Recognize index notation:

Consider the factors of 360.

$$360 = 2 \times 2 \times 2 \times 3 \times 3 \times 5$$

Here, factor 2 is 3 times = 2^3 (index = 3)

factor 3 is 2 times = 3^2 (index = 2)

and factor 5 is 1 time = 5^1 (index = 1)

So, prime factorization of 360 can be expressed as

$$360 = 2^3 \times 3^2 \times 5^1$$

The prime factorization of 360 in the above form is known as index notation.

2	360
2	180
2	90
3	45
3	15
5	5
	1

Example: Find prime factorization and express it in index notation.

(i) 432 (ii) 1500

Solution:

(i) 432

2	432
2	216
2	108
2	54
3	27
3	9
3	3
	1

Hence,

$$432 = 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 3$$

$$= 2^4 \times 3^3$$

(ii) 1500

2	1500
2	750
3	375
5	125
5	25
5	5
	1

Hence,

$$1500 = 2 \times 2 \times 3 \times 5 \times 5 \times 5 = 2^2$$

$$\times 3^1 \times 5^3$$

EXERCISE 3.3

Q1. Write in index notation.

(i) $2 \times 2 \times 2 \times 3 \times 5 \times 3$

(ii) $3 \times 5 \times 5 \times 5 \times 7 \times 7$

(iii) $2 \times 3 \times 3 \times 7 \times 7 \times 3 \times 3$

(iv) $4 \times 4 \times 4 \times 2 \times 2 \times 5$

(v) $6 \times 6 \times 3 \times 3 \times 3 \times 2 \times 2 \times 2$

(vi) $8 \times 8 \times 8 \times 3 \times 3 \times 3 \times 2$

Q2. Find the prime factorization by division method and express in index notation.

(i) 24

(ii) 48

(iii) 216

(iv) 250

(v) 469

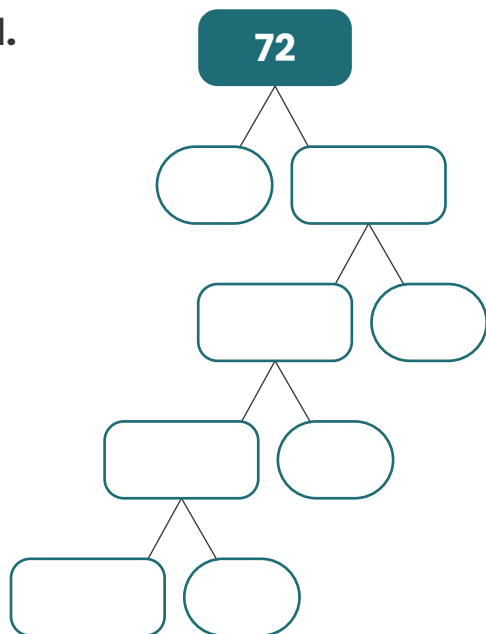
(vi) 540

(vii) 1024

(viii) 5000

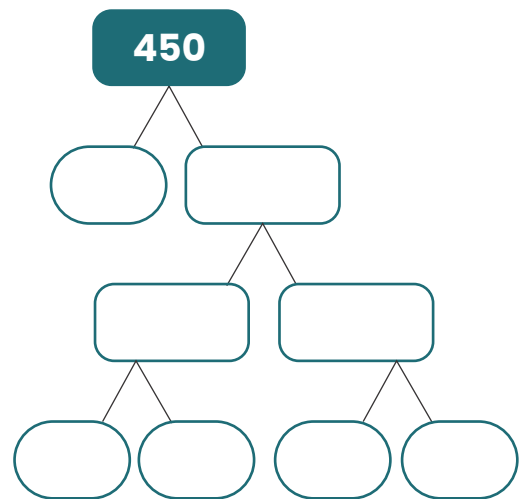
Q3. Find the prime factorization in the factor trees.

1.



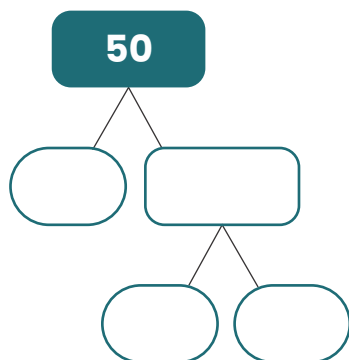
72= _____

2.



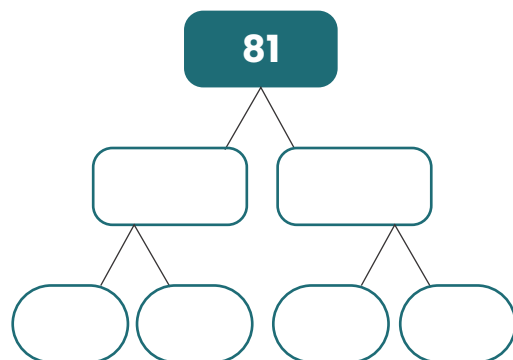
450= _____

3.



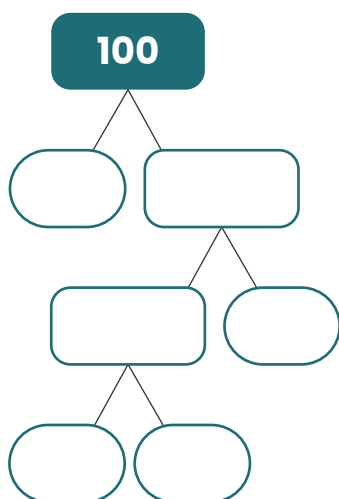
50= _____

4.



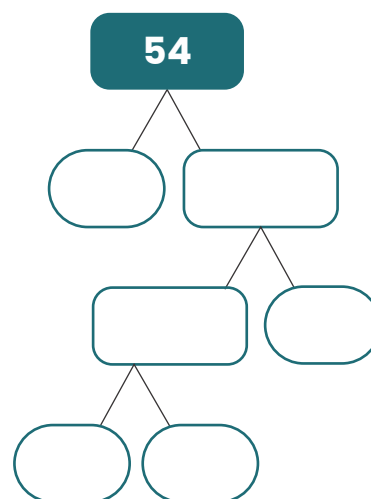
81= _____

5.



$100 = \underline{\hspace{2cm}}$

6.



$54 = \underline{\hspace{2cm}}$

Q4. Find the prime factors of each number by using Prime Factor tree.

15 15 = <u> </u>	21 21 = <u> </u>	25 25 = <u> </u>
35 35 = <u> </u>	49 49 = <u> </u>	51 51 = <u> </u>

55 55= _____	57 57= _____	65 65= _____
--	--	--

HCF (HIGHEST COMMON FACTOR)

HCF of two or more given numbers is the common factor of the given numbers which is the greatest of all the common factors.

There are two main methods to find HCF:

(i) Prime Factorization method

(ii) Long division method

(i) Prime Factorization Method:

Example: Find HCF of 48, 36 and 40.

Solution:

2 48 2 24 2 12 2 6 3 3 1	2 36 2 18 3 9 3 3 1	2 40 2 20 2 10 5 5 1
---	---	--

Here 48 = 2 x 2 x 2 x 2 x 3
 36 = 2 x 2 x 3 x 3
 40 = 2 x 2 x 2 x 5

Common factors are 2 and 2

So, HCF = 2 x 2 = 4

(ii) Long Division Method:**Example 1:** Find HCF of 24 and 64 by long division method.**Solution:**

$$\begin{array}{r}
 24 \overline{) 64} \quad (2 \\
 \underline{- 48} \\
 16 \overline{) 24} \quad (1 \\
 \underline{- 16} \\
 \text{HCF} \longrightarrow 8 \overline{) 16} \quad (2 \\
 \underline{- 16} \\
 0
 \end{array}$$

So, HCF = 8

Example 2: Find HCF of 16, 36 and 70.**Solution:**

$$\begin{array}{r}
 16 \overline{) 36} \quad (2 \\
 \underline{- 32} \\
 4 \overline{) 16} \quad (4 \\
 \underline{- 16} \\
 0
 \end{array}$$

So, HFC of 16 and 36, = 4

Now we find HFC of 4 and 70

$$\begin{array}{r}
 4 \overline{) 70} \quad (17 \\
 \underline{- 68} \\
 20 \\
 \underline{- 16} \\
 4 \\
 \text{HCF} \longrightarrow 2 \overline{) 4} \quad (2 \\
 \underline{- 4} \\
 0
 \end{array}$$

Hence, the HCF of 16, 36 and 70 is 2.

EXERCISE 3.4

Q1. Find the HCF of the following numbers by prime factorization method.

(i) 50, 75

(ii) 98, 196

(iii) 144, 198

(iv) 120, 144, 204

(v) 106, 159, 265

(vi) 12, 48, 36, 24

(vii) 60, 70, 420, 480

Q2. Find HCF of the following numbers by long division method.

(i) 12, 20

(ii) 81, 117

(iii) 935, 1320

(iv) 252, 576

(v) 2241, 8217, 747

(vi) 30, 120, 90, 210

Q3. Find the highest common factor of the following numbers.

1. HCF of 12 and 15 _____

5. HCF of 16 and 24 _____

2. HCF of 10 and 8 _____

6. HCF of 12 and 18 _____

3. HCF of 20 and 15 _____

7. HCF of 50 and 15 _____

4. HCF of 25 and 10 _____

8. HCF of 20 and 40 _____

Q4. Find the HCF of the following through prime factorization.

1. 9, 12

2. 24, 28

9: _____

24: _____

12: _____

28: _____

HCF: _____

HCF: _____

3. 63, 72

63: _____

72: _____

HCF: _____

4. 36, 84

36: _____

84: _____

HCF: _____

5. 13, 39, 52

13: _____

39: _____

52: _____

HCF: _____

6. 30, 45, 60

30: _____

45: _____

60: _____

HCF: _____

LCM (LEAST COMMON MULTIPLE)

LCM of two or more given numbers is the smallest of all their common multiples.

There are two main methods to find LCM:

(i) Prime Factorization method

(ii) Long division method

(i) Prime Factorization Method:

Example: Find LCM of 12, 18, 24.

Solution:

2	12
2	6
3	3
	1

2	18
3	9
3	3
	1

2	24
2	12
2	6
3	3
	1

Here

$$12 = 2 \times 2 \times 3$$

$$18 = 2 \times 3 \times 3$$

$$24 = 2 \times 2 \times 2 \times 3$$

Common factors are 2, 2, 3

Non-common factors are: 2, 3

$$\begin{aligned}\text{So, LCM} &= 2 \times 2 \times 3 \times 2 \times 3 \\ &= 72\end{aligned}$$

(ii) Long Division Method:

Example 1:

Find LCM of 3, 4, 5.

Solution:

2	3, 4, 5
2	3, 2, 5
3	3, 1, 5
5	1, 1, 5
	1, 1, 1

$$\begin{aligned}\text{Here LCM} &= 2 \times 2 \times 3 \times 5 \\ &= 60\end{aligned}$$

Example 2:

Find LCM of 12, 15, 50.

Solution:

2	12, 15, 50
2	6, 15, 25
3	3, 15, 25
5	1, 5, 25
5	1, 1, 5
	1, 1, 1

$$\begin{aligned}\text{Here LCM} &= 2 \times 2 \times 3 \times 5 \times 5 \\ &= 300\end{aligned}$$

EXERCISE 3.5

Q1. Using prime factorization, find the LCM of the following numbers.

- (i) 12, 25 and 40 (ii) 21, 49 and 63 (iii) 144, 180 and 384
 (iv) 108, 135 and 162 (v) 35, 65 and 75 (vi) 24, 36, 48 and 72

Q2. Calculate the LCM of the following numbers by division method.

- (i) 45 and 55 (ii) 21, 35 and 70
 (iii) 24, 40 and 60 (iv) 72, 108 and 120

Q3. Find LCM of the following.

	Question	LCM
1)	6, 4	
2)	3, 5	
3)	7, 2	
4)	8, 4	
5)	10, 3	
6)	5, 7	

Q4. Find the least common multiple of each pair of numbers.

	Pair	LCM
1)	9, 15	_____
2)	4, 8	_____
3)	18, 3	_____
4)	22, 6	_____

Q5. List the first five multiples of the following numbers.

- a.** 2 **b.** 6 **c.** 3 **d.** 10
e. 25 **f.** 12 **g.** 7 **h.** 13

Q6. Find the lowest common multiples of the following groups of numbers.

- a.** 4, 6 and 12 **b.** 2, 7 and 8 **c.** 5, 6 and 9
d. 4, 5 and 6 **e.** 3, 5 and 6 **f.** 2, 5 and 15

APPLICATIONS OF HCF AND LCM

Example 1: Find the greatest length of a measuring tape which can be used to measure exactly 520 cm and 360 cm.

Solution: The greatest length of the tape is the HCF of 520 cm and 360 cm.

Let's find the required HCF by the division method.

$$\begin{array}{r}
 360 \overline{) 520} \quad (1 \\
 \underline{-360} \\
 160 \\
 160 \overline{) 360} \quad (2 \\
 \underline{-320} \\
 40 \\
 40 \overline{) 160} \quad (4 \\
 \underline{-160} \\
 0
 \end{array}$$

The HCF of 520 and 360 is 40.

Hence, the required length of the tape is 40 cm.

Example 2: Find the greatest length of wooden scale which can be used to measure 63 cm, 85 cm and 47 cm leaving remainders 3 cm, 5 cm and 7 cm respectively.

Solution:

63, 85, 47
<u>-3, -5, -7</u>
60, 80, 40

Here, we have to find HCF of 60, 80, and 40.

Let's find the HCF by division method.

$$\begin{array}{r}
 60 \overline{) 80} \quad (1 \\
 \underline{-60} \\
 20 \\
 20 \overline{) 60} \quad (2 \\
 \underline{-60} \\
 0
 \end{array}$$

As HCF of 60 and 80 is 20.
Now we find HCF of 20 and 40.

$$\begin{array}{r} 20 \overline{) 40} \quad 2 \\ \underline{-40} \\ 0 \end{array}$$

As the HCF of 60, 80, 40 is 20.
So, the required length of the wooden scale is 20 cm.

Example 3: Find the least quantity of sugar which can be exactly measured by 100 gm, 150 gm or 200 gm weights.

Solution:

The required least quantity of sugar can be found with the help of LCM.

2	100, 150, 200
2	50, 75, 100
2	25, 75, 50
3	25, 75, 25
5	25, 25, 25
5	5, 5, 5
	1, 1, 1

Here $LCM = 2 \times 2 \times 2 \times 3 \times 5 \times 5 = 600$

Hence the required quantity of sugar is 600 gm.

WORD PROBLEM

- 1.** Find the greatest length of a wooden scale which can be used to measure 540 cm and 360 cm exactly.
- 2.** Find the greatest number which divides 232 and 305 leaving 7 and 5, respectively, as remainder.
- 3.** Find the largest number which divides 245 and 1029 leaving remainder 5 in each case.
- 4.** The length, breadth and height of a room are 8 m, 6 m and 4 m respectively. Determine the longest tape which can measure the three dimensions of the room exactly.
- 5.** Three bells ring at an interval of 5, 10 and 15 minutes respectively. If these started ringing at 8 A.M., find the time when they will again ring together.
- 6.** Find the least number which when divided by 30, 45 and 60 leaves 9 as remainder.